



Caringbah High School

2016

Trial HSC Examination

Mathematics Extension I

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen Black pen is preferred
- Board-approved calculators may be used
- A data sheet is provided at the back of this paper
- In Questions 11–15, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I 10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II 60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 4 The remainder when $P(x) = 2x^3 - 6x^2 + 4x + 3$ is divided by $2x - 1$ is
- A) 3
B) -9
C) $3\frac{3}{4}$
D) $\frac{-3}{4}$
- 5 The exact value of $\cos 15^\circ$ is
- A) $\frac{\sqrt{6} + \sqrt{2}}{4}$
B) $\frac{\sqrt{3}}{4}$
C) $\frac{1}{4}$
D) $\frac{\sqrt{6} - \sqrt{2}}{4}$
- 6 $\int 2 \cos^2 x \, dx = v$
- A) $-\sin x \cos x + x + c$
B) $\frac{1}{2} \sin 2x + x + c$
C) $\frac{2}{3} \cos^3 x + c$
D) $\frac{-2}{\sqrt{1-x^2}} + c$
- 7 The velocity of a particle at a position x is given by $\dot{x} = 2e^{\frac{-x}{2}}$ m/s. The particles acceleration when its displacement is -2 metres is
- A) $-e \text{ m/s}^2$
B) $\frac{-4}{e^2} \text{ m/s}^2$
C) $-2e^2 \text{ m/s}^2$
D) $e^2 \text{ m/s}^2$
- 8 The value of $\sin^{-1}(\frac{1}{2}) + \cos^{-1}(\frac{-\sqrt{3}}{2})$ is
- A) $\pi + \frac{\pi}{3}$
B) 0
C) $\frac{7\pi}{6}$
D) π

9 If $\log_a x = p$ and $\log_a y = q$, find the value of $\log_a x^2 y$ in terms of p and q .

A) $p^2 q$

B) $2p + q$

C) $p^2 + q$

D) $q - 2p$

10 When $y = e^{x+2}$ is rotated about the y axis between $x = 0$ and $x = 2$, its volume is given by

A) $\pi \int_{e^2}^{e^4} (\ln y - 2)^2 dy$

B) $\pi \int_{e^2}^{e^4} e^{2x+4} dx$

C) $\pi \int_0^2 e^{x+2} dx$

D) $\pi \int_0^2 (\ln y - 2) dy$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- a) Find the gradient of the tangent to the curve $y = \cos^3 x$ at $x = \frac{\pi}{6}$ 2
- b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ 1
- c) Consider the function $f(x) = 3 \sin^{-1}\left(\frac{x}{2}\right)$
- i) Find the value of $f(2)$ 1
- ii) State the domain and range of this function 2
- iii) Draw the graph of $y = f(x)$ 2
- iv) Find $f'(x)$ 1
- d) A particle moves in a straight line so its position x from a fixed point O at time t is given by $x = 3 \sin 2t + 4 \cos 2t$.
- i) If the motion is expressed in the form $x = r \sin(2t + \alpha)$ find the value of the constants r and α . (α to the nearest degree) 2
- ii) Show the motion is simple harmonic. 2
- iii) What is the period of the oscillation? 1
- iv) Determine the maximum displacement from the centre of the motion. 1

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- a) Write a primitive for $(5-2x)^4$ 2
- b) Find $\int \tan x \, dx$ 2
- c) If α, β, γ are the roots of the equation $x^3 - 4x + 1 = 0$ evaluate $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2
- d) A bottle of lemonade is taken out of a fridge at $4^\circ C$ into a room where the air temperature is $25^\circ C$. The rate at which the lemonade warms follows Newton's law, that is $\frac{dT}{dt} = k(T - 25)$ where $k < 0$, time t is measured in minutes, and the temperature T is in degrees celsius.
- i) Show that $T = 25 + Ae^{kt}$ is a solution to $\frac{dT}{dt} = k(T - 25)$ and find the value of A 2
- ii) The temperature of the lemonade reaches $15^\circ C$ in 45 minutes. Find the value of k to four decimal places. 2
- iii) Find the temperature of the lemonade 90 minutes after being removed from the fridge, to nearest degree. 1
- e) Prove by the method of mathematical induction that $\sum_{r=1}^n 5^{r-1} = \frac{5^n - 1}{4}$ 4

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

a) Find the first derivative of $y = \log_e \left(\frac{1}{\sqrt{\cos x}} \right)$ 3

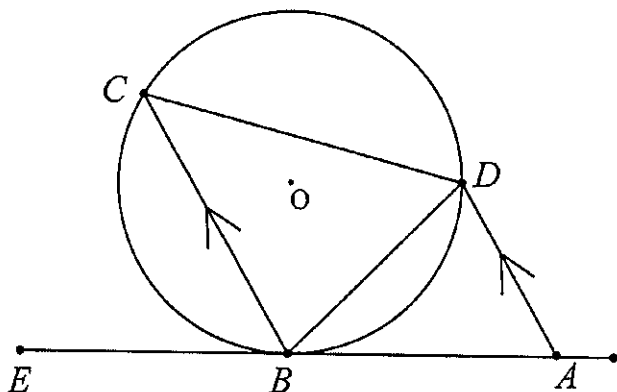
b) A capsule is in the shape of a cylinder with a hemisphere on each end. The radius of the cylindrical section is r cm, and the volume of the capsule is 16 cm^3 .

i) If the height of the cylinder is 4 cm show that $r^3 + 3r^2 = \frac{12}{\pi}$ 2

ii) Show that one solution of the equation $r^3 + 3r^2 = \frac{12}{\pi}$ lies between $r = 0$ and $r = 1$ 1

iii) The equation $r^3 + 3r^2 = \frac{12}{\pi}$ has one root close to $r = 0.9$. Use one application of Newton's method of approximation to give a better approximation to three decimal places. 2

c) AE is tangent at B and $AD \parallel BC$. Prove that $\triangle BCD \parallel \triangle DBA$ 2



d) Find the indefinite integral of $\int \frac{1}{\sqrt{1-4x^2}} dx$ 2

e) The polynomial $3x^3 - 17x^2 - 8x + 12 = 0$ has roots α, β, γ . Given that the product of two of the roots is 4, solve the equation for α, β, γ 3

End of Question 13

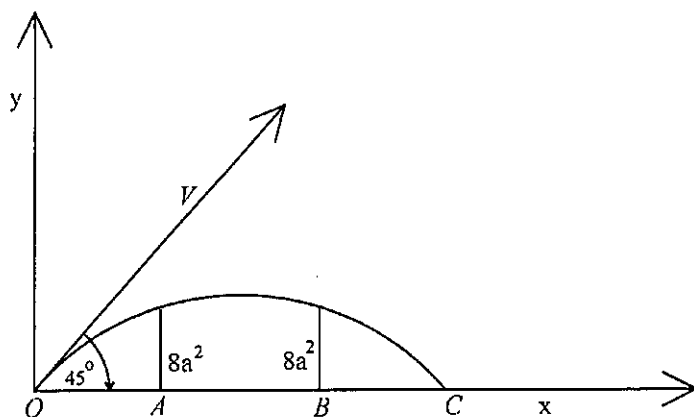
Question 14 (15 marks) Use a SEPARATE writing booklet.

- a) Evaluate $\int_0^{\sqrt{3}} \frac{x}{\sqrt{1+x^2}} dx$ using the substitution $u = 1+x^2$ 3
- b) i) Sketch the graph of $y = \cos x$ and $y = \sin x$ on the same diagram for $0 \leq x \leq \frac{\pi}{2}$ 1
- ii) Show that if $0 < x < \frac{\pi}{4}$, then $\sin 2x > 2 \sin^2 x$ 2

Question 14 continues on page 10

Question 14 (continued)

- c) A projectile fired with velocity V and at an angle of 45° to the horizontal, just clears the top of two vertical posts of height $8a^2$ units. The posts at A and B are $12a^2$ units apart. Also $OA=BC=b$ units. There is no air resistance and the acceleration due to gravity is g .



If the projectile is at a point $P(x, y)$ at time t , expressions for x and y in terms of t are $x = \frac{Vt}{\sqrt{2}}$ and $y = \frac{-gt^2}{2} + \frac{Vt}{\sqrt{2}}$. **Do not prove these results.**

- i) Show that the path of the projectile is given by $y = x - \frac{gx^2}{V^2}$ 2
- ii) Using the information in (ii) show that the range of the projectile is $\frac{V^2}{g}$ 2
- iii) If the first post is b units from the origin show that
 - (α) $\frac{V^2}{g} = 2b + 12a^2$ 1
 - (β) $8a^2 = b - \frac{gb^2}{V^2}$ 1
- iv) Hence or otherwise prove that $V = 6a\sqrt{g}$. 3

End of Exam

SOLUTIONS

Multiple Choice

1, C 2, B 3, B 4, C 5, A 6, B 7, C 8, D 9, B 10, A

Question 11

(a) $y = \cos^3 x$

$$y' = 3(\cos x)^2 \cdot -\sin x$$

when $x = \pi/6$

$$\begin{aligned} M_T &= 3(\cos \pi/6)^2 \cdot -\sin \pi/6 \\ &= 3(\sqrt{3}/2)^2 \cdot -1/2 \\ &= -9/8 \end{aligned}$$

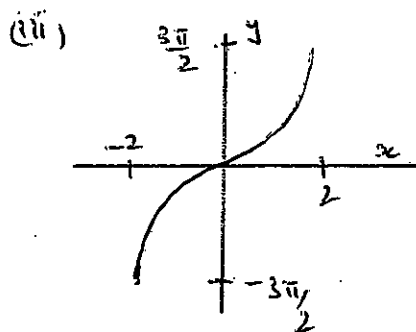
(b) $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$

$$\begin{aligned} &= \frac{\sin 3x}{3x} \times \frac{3}{5} \\ &= 3/5 \end{aligned}$$

(c) (i) $f(x) = 3\pi/2$

(ii) D: $-2 \leq x \leq 2$

R: $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$



(iv) $f(x) = 3\sin^{-1}(x/2)$

$$\begin{aligned} f'(x) &= 3 \cdot \frac{1}{\sqrt{1-(x/2)^2}} \times \frac{1}{2} \\ &= \frac{3}{2} \left[\frac{2}{\sqrt{4-x^2}} \right] \\ &= \frac{3}{\sqrt{4-x^2}} \end{aligned}$$

(d) Question 11 (cont'd)

(i) $r = 5, \alpha = 53^\circ$

(ii) $x = 5 \sin(2t + 53^\circ)$

$$\begin{aligned} \dot{x} &= 5 \cos(2t + 53^\circ) \times 2 \\ &= 10 \cos(2t + 53^\circ) \end{aligned}$$

$$\ddot{x} = -20 \sin(2t + 53^\circ)$$

$$\ddot{x} = -4x$$

(iii) $T = \frac{2\pi}{2}$

$$= \pi$$

(iv) 5

Question 12

(a) $\frac{(5-2x)^5}{-10} + C$

(b) $\int \tan x \, dx$

$$\begin{aligned} &= \int \frac{\sin x}{\cos x} \, dx \\ &= -\int \frac{\sin x}{\cos x} \\ &= -\ln(\cos x) + C \end{aligned}$$

(c) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$

$$\begin{aligned} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{-4}{-1} \\ &= +4 \end{aligned}$$

(d) (i) $T = 25 + Ae^{kt}$

$$\frac{dT}{dt} = k \cdot Ae^{kt}$$

$$= k(T - 25)$$

when $t = 0$, $T = 4$

$$4 = 25 + Ae^0$$

$$A = -21$$

$$\therefore T = 25 - 21e^{kt}$$

(ii) $15 = 25 - 21e^{45K}$

$$\frac{10}{21} = e^{45K}$$

$$\ln\left(\frac{10}{21}\right) = 45K$$

$$K = -0.0165$$

(iii) $T = 25 - 21e^{-0.0165(90)}$

$$T = 20^\circ \text{C (to nearest degree)}$$

(e) $\sum_{r=1}^n 5^{r-1} = \frac{5^n - 1}{4}$

when $r=1$ $5^{r-1} = \frac{5^1 - 1}{4}$
 $1 = 1$

Assume true for $n=k$

$$S_k = \frac{5^k - 1}{4}$$

Prove true for $n=k+1$

$$\begin{aligned} S_{k+1} &= S_k + T_{k+1} \\ &= \frac{5^k - 1}{4} + 5^k \\ &= \frac{5^k - 1 + 4 \cdot 5^k}{4} \\ &= \frac{5 \cdot 5^k - 1}{4} \\ &= \frac{5^{k+1} - 1}{4} \quad (\text{plus}) \\ &\quad (\text{Statement}) \end{aligned}$$

(a) $y = \log_e \left(\frac{1}{\sqrt{\cos x}} \right)$

$$y = \log_e [(\cos x)^{-1/2}]$$

$$y' = \frac{-\frac{1}{2}(\cos x)^{-3/2} \cdot -\sin x}{(\cos x)^{-1/2}}$$

$$\begin{aligned} y' &= \frac{1}{2} \sin x (\cos x)^{-1} \\ &= \frac{1}{2} \frac{\sin x}{\cos x} \\ &= \frac{1}{2} \tan x \end{aligned}$$

(b) $V = \pi r^2 \cdot 4 + \frac{4}{3} \pi r^3$

(i)

$$16 = 4\pi r^2 + \frac{4}{3}\pi r^3$$

$$48 = 12\pi r^2 + 4\pi r^3$$

$$48 = 4\pi (3r^2 + r^3)$$

$$\frac{12}{\pi} = r^3 + 3r^2$$

(ii)

$$\text{Let } f(r) = r^3 + 3r^2 - \frac{12}{\pi}$$

$$r=0 \quad f(0) = -\frac{12}{\pi}$$

$$\begin{aligned} r=1 \quad f(1) &= 1 + 3 - \frac{12}{\pi} \\ &= 0.18 \dots \end{aligned}$$

$$f(0) < 0 \text{ and } f(1) > 0$$

iii) $f(r) = r^3 + 3r^2 - \frac{12}{\pi}$

$$f(0.9) \doteq -0.6607$$

$$f'(r) = 3r^2 + 6r$$

$$f'(0.9) = 7.83$$

$$r_2 = r_1 - \frac{f(r_1)}{f'(r_1)}$$

$$r_2 \doteq 0.984$$

Question 13

(c)

$$\angle ADB = \angle DBC$$

(alt $\angle$'s // lines
CB, DA)

$$\angle DBA = \angle DCB$$

(alt seg thm)

$$\triangle BCD \parallel \triangle DBA$$

(equiangular)

$$(d) \int \frac{1}{\sqrt{1-4x^2}} dx$$

$$= \int \frac{1}{\sqrt{4\left[\left(\frac{1}{2}\right)^2 - x^2\right]}} dx$$

$$= \frac{1}{2} \int \frac{1}{\sqrt{\left(\frac{1}{2}\right)^2 - x^2}}$$

$$= \frac{1}{2} \sin^{-1}\left(\frac{x}{\frac{1}{2}}\right) + C$$

$$= \frac{1}{2} \sin^{-1} 2x + C$$

$$(e) \alpha + \beta + \gamma = \frac{17}{3} \quad \text{--- (1)}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{8}{3} \quad \text{--- (2)}$$

$$\alpha\beta\gamma = -4 \quad \text{--- (3)}$$

$$\text{Let } \alpha\beta = 4 \Rightarrow \beta = \frac{4}{\alpha}$$

$$4\gamma = -4$$

$$\gamma = -1$$

using (1)

$$\alpha + \beta - 1 = \frac{17}{3}$$

$$\alpha + \beta = \frac{20}{3}$$

$$\text{sub } \beta = \frac{4}{\alpha}$$

$$\alpha + \frac{4}{\alpha} = \frac{20}{3}$$

$$3\alpha^2 - 20\alpha + 12 = 0$$

$$(\alpha - 6)(3\alpha - 2) = 0$$

$$\alpha = 6, \alpha = \frac{2}{3}$$

$\therefore$ Roots are

$$6, \frac{2}{3}, -1$$

Question 14

$$(a) \int_0^{\sqrt{3}} \frac{x dx}{\sqrt{1+x^2}}$$

$$\text{If } u = 1+x^2$$

$$du = 2x \cdot dx$$

$$\text{when } x = \sqrt{3}$$

$$u = 4$$

$$\text{when } x = 0$$

$$u = 1$$

$$\frac{1}{2} \int_0^{\sqrt{3}} \frac{2x}{\sqrt{1+x^2}} dx$$

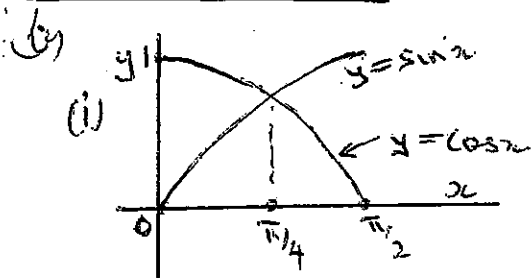
$$\frac{1}{2} \int_1^4 \frac{du}{u^{1/2}}$$

$$\frac{1}{2} \int_1^4 u^{-1/2} du$$

$$\frac{1}{2} \left[\frac{u^{1/2}}{1/2} \right]_1^4$$

$$= \frac{1}{2} [4 - 2]$$

$$= 1$$



for $0 < x < \pi/4$

(ii) $\cos x > \sin x$ and

$\sin x \sin x > 0$

$$\cos x \sin x > \sin^2 x$$

$$2 \cos x \sin x > 2 \sin^2 x$$

$$\sin 2x > 2 \sin^2 x$$

OR $\sin 2x > 2 \sin^2 x$ " "

$$2 \sin x \cos x > 2 \sin^2 x$$

$\cos x > \sin x$ (since $\sin x > 0$)

graph shows $\cos x$ is higher than $\sin x$ for

$$0 < x < \pi/4$$

(C) (i) $x = \frac{vt}{\sqrt{2}}$, $y = -\frac{gt^2}{2} + \frac{vt}{\sqrt{2}}$

$$t = \frac{\sqrt{2}x}{v}$$

$$\therefore y = -\frac{g}{2} \left(\frac{\sqrt{2}x}{v} \right)^2 + \frac{v}{\sqrt{2}} \left(\frac{\sqrt{2}x}{v} \right)$$

$$y = -\frac{g}{2} \cdot \frac{2x^2}{v^2} + x$$

$$y = -\frac{gx^2}{v^2} + x$$

(ii) when $y = 0$

$$0 = -\frac{gx^2}{v^2} + x$$

$$0 = x \left(1 - \frac{gx}{v^2} \right) = 0$$

start
 $x = 0$

$$1 - \frac{gx}{v^2} = 0$$

$$\frac{gx}{v^2} = 1$$

$$x = \frac{v^2}{g}$$

(ii) (A) $OC = \frac{v^2}{g} = OA + AB + BC$

$$\frac{v^2}{g} = b + 12a^2 + b$$

$$\frac{v^2}{g} = 2b + 12a^2$$

$$y = x - \frac{gx^2}{v^2}$$

(B) $x = b$, $y = 8a^2$

$$8a^2 = b - \frac{gb^2}{v^2}$$

Question 14

$$(IV) \frac{V^2}{g} = 2b + 12a^2 \quad - (1)$$

$$8a^2 = b - \frac{gb^2}{V^2} \quad - (2)$$

from (1)

$$V^2 = g(2b + 12a^2)$$

sub into (2)

$$8a^2 = b - \frac{gb^2}{g(2b + 12a^2)}$$

$$8a^2 = \frac{2b^2 + 12a^2b - b^2}{2b + 12a^2}$$

$$16a^2b + 96a^4 = b^2 + 12a^2b$$

$$b^2 - 4a^2b - 96a^4 = 0$$

$$(b - 12a^2)(b + 8a^2) = 0$$

$$b = 12a^2 \quad b = -8a^2$$

$$\therefore b = 12a^2$$

$$\frac{V^2}{g} = 2(12a^2) + 12a^2$$

$$\frac{V^2}{g} = 36a^2$$

$$V = 6a\sqrt{g}$$

OR (IV)

$$\frac{V^2}{g} = 2b + 12a^2 \quad - (1)$$

$$8a^2 = b - \frac{gb^2}{V^2} \quad - (2)$$

$$\text{from (1)} \quad b = \frac{V^2}{2g} - 6a^2$$

sub into (2)

$$8a^2 = \frac{V^2}{2g} - 6a^2 - \frac{g}{V^2} \left(\frac{V^2}{2g} - 6a^2 \right)^2$$

$$8a^2 = \left(\frac{V^2}{2g} - 6a^2 \right) \left[1 - \frac{g}{V^2} \left(\frac{V^2}{2g} - 6a^2 \right) \right]$$

$$8a^2 = \left(\frac{V^2}{2g} - 6a^2 \right) \left[1 - \frac{1}{2} + \frac{6a^2g}{V^2} \right]$$

$$8a^2 = \frac{V^2 - 12a^2g}{2g} \left[\frac{1}{2} + \frac{6a^2g}{V^2} \right]$$

$$8a^2 = \left(\frac{V^2 - 12a^2g}{2g} \right) \left[\frac{V^2 + 12a^2g}{2V^2} \right]$$

$$8a^2 = \frac{V^4 - 144a^4g^2}{4gV^2}$$

$$32a^2gV^2 = V^4 - 144a^4g^2$$

$$V^4 - 32a^2gV^2 - 144a^4g^2 = 0$$

$$(V^2 - 36a^2g)(V^2 + 4a^2g) = 0$$

$$\therefore V^2 - 36a^2g = 0$$

$$V^2 = 36a^2g$$

$$V = 6a\sqrt{g}$$